

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2023

Course: T.E.

Branch : Mechanical Engineering

Semester :VI

Subject Code & Name: BTMC601/Manufacturing Processes -II

Max Marks: 60

Date: 13 July 2023

Duration: 3 Hrs.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) What is grinding? Explain various types of grinding wheels with suitable sketch.	Apply	6
B) Explain Design Consideration for Grinding.	Understanding	6
C) What is grinding wheel wear? Explain the mechanisms of wheel wear.	Apply	6
Q.2 Solve Any Two of the following.		12
A) Explain Merchant's Circle Diagram.	Apply	6
B) Describe various angles of single point cutting tool showing angles on figure.	Understanding	6
C) Differentiate between Orthogonal and Oblique Cutting.	Analyze	6
Q. 3 Solve Any Two of the following.		12
A) What is meant by tool life? Which factors affects tool life?	Apply	6
B) Write short notes on – (i) Cubic Boron (ii) Nitride Coated Tools	Remember	6
C) Explain properties and types of cutting fluid.	Remember	6
Q.4 Solve Any Two of the following.		12
A) State limitations and advantage of Powder Metallurgy process.	Apply	6
B) What is sintering? Explain three stages of sintering.	Understanding	6
C) Explain mechanical method of powder production.	Apply	6
Q. 5 Solve Any Two of the following.		12
A) Discuss some of the defects that can occur in plastic injection molding.	Evaluate	6
B) What are the different strengthening techniques of glass? Explain any 2 in details	Apply	6
C) What is (a) Parison, (b) Plastisol, and (c) Prepreg?	Understanding	6

***** End *****